Akash Verma

github.com/Ak0801 akash.verma6712@gmail.com https://akash.geekfleet.dev/ +91 93066 20996

Education

Indian Institute of Science Education and Research Mohali, Punjab, India Aug'20 – May'25
BS–MS Data Science and Chemistry *CPI: 7.8*

Skills

Programming Languages: Python
Generative AI & LLMs: Azure OpenAI (GPT Models), Hugging Face Transformers, Prompt Engineering, LLM Applications
RAG & Agentic AI: LangChain, RAG (Retrieval-Augmented Generation), Vector Search, MCP (Model Context Protocol), Agentic AI Workflows
AI/ML & Deep Learning: CNN, ANN, PyTorch, TensorFlow, Transfer Learning, Model Evaluation Metrics
Computer Vision: OpenCV, YOLO, ArUco Markers, K-means Clustering, Contour Analysis
Data Engineering & Pipelines: Data Cleaning, Preprocessing, End-to-End AI Pipelines, Model Deployment Workflows, Kafka, AWS SQS
Vector Databases & Retrieval: Redis (Vector Similarity Search), Similarity Search Techniques
Databases: PostgreSQL, SQL, NoSQL
Cloud Infrastructure: Microsoft Azure, AWS
Embedded Systems: Raspberry Pi, NVIDIA Jetson, GPIO Interfacing

Work Experience

Techdome - Indore, Madhya Pradesh — *Associate Data Science Engineer II* June'25 – Jan'26

- Designed and implemented an end-to-end AI video generation platform by integrating multiple generative AI models, APIs, and automated workflows.
- Built scalable AI pipelines for video synthesis, voice generation, and content orchestration using Python and cloud-based services.
- Developed a multi-camera person re-identification system using YOLO and Deep SORT, enabling consistent identity tracking across video feeds.
- Optimized feature matching using cosine similarity and color histogram correlation to improve re-identification accuracy in real-world environments.

Infinity Labs - Greater Noida, Uttar Pradesh — *Data Science Engineer Intern* June'23 – Aug'23

- Developed a facial recognition system for automated attendance using OpenCV, enabling real-time face detection and identification.
- Implemented database-backed attendance logging using SQLite and integrated a Flask-based web dashboard for reporting.
- Improved system reliability by optimizing image preprocessing and handling concurrent detection and database updates.
- Deployed and tested the complete solution on Raspberry Pi, ensuring efficient performance on edge devices.

Projects and Experience

AI Video Generation Tool

- Built end-to-end AI Video generation pipeline using mutiple generative AI models and API's
- Developed comprehensive AI model integration framework with voice synthesis capabilities
- Engineered muti-step AI workflow orchestration with intelligent task scheduling
- Architected scalable AI content pipeline with AWS cloud infrastructure

Person Re-Identification and Tracking System

- Developed a multi-camera person tracking system using YOLO for detection and Deep SORT for real-time re-identification.
- Fused cosine similarity and HSV histogram correlation for accurate Global ID (GID) assignment across multiple videos.
- Implemented homography-based 2D mapping to visualize presence on a site blueprint.

Automated Bullet Dimension Extraction Using Computer Vision —

- Developed a YOLO-based deep learning model to automate bullet detection and classification for forensic applications.
- Built a workflow combining K-means clustering, grayscale conversion, and contour detection for accurate bullet segmentation.
- Used ArUco marker–based camera calibration to map pixels to real-world units for dimensional measurements.
- Programmed Raspberry Pi GPIO to control linear actuators for precision mechanical movements.
- Integrated motor drivers for efficient stepper motor control in robotics and automation projects.
- Outlined future enhancements, including real-time processing on Raspberry Pi and high-resolution image capture.

Facial Recognition–Based Attendance System —

- Built a facial recognition attendance system with OpenCV for real-time face detection and recognition.
- Integrated SQLite for profile and attendance storage.
- Created a Flask web interface for real-time attendance reports.
- Optimized for reliability, handling simultaneous detection and database updates with minimal latency.
- Deployed the system on a Raspberry Pi for a compact IoT setup.

Achievements

Appreciation Award: Appreciation Award for outstanding performance
Internship Certificate: Work Experience Certificate from Infinity Labs, Noida
Finlatics: Data Science Certification Course

Hobbies

- **Football** — Winner of Inter-Batch Sports Tournament
- **Swimming**
- **Trekking**